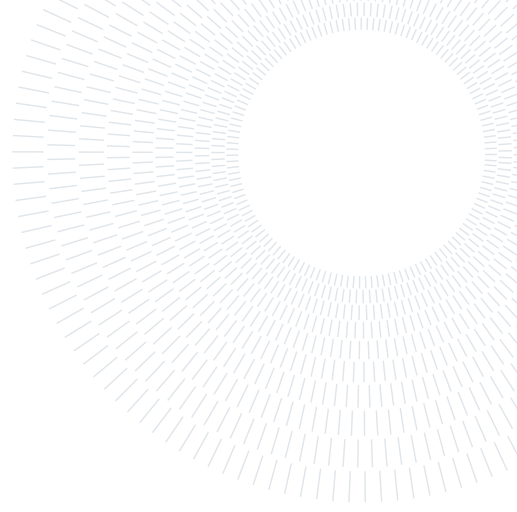




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Numerical Results Report of the First Assignment

AERONAUTICAL ENGINEERING - STRUCTURAL DYNAMICS AND AEROELASTICITY COURSE

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In the next pages, after a quick explanation of which assumptions and choices were made, the numerical results obtained from the code will be presented.

All the results in the report are referred to the chordwise position x of the strut fixed at $x = -0.40$ m.

First plot (figure 1) represents the divergence dynamic pressure as function of the chordwise position of the strut.

The discontinuity of the plot between $x \approx 0.35$ m and $x \approx -0.15$ m means that in this interval the strut makes the structure intrinsically stable against the divergence.

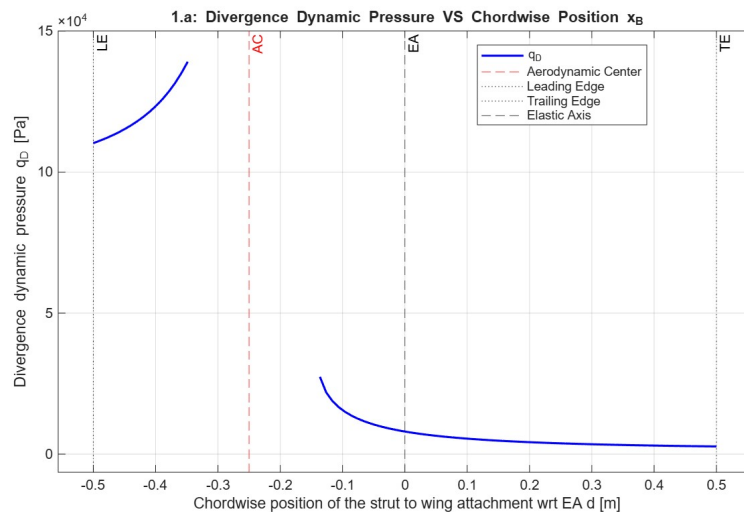


Figure 1: q_D VS position of the strut along the chord

Figure 2 shows that reversal condition is reached when the control effectiveness drops to 0.

Third plot (figure 3) shows how the strut has an influence on the bending and torsional behaviour of the wing. The strut has a very positive effect on reducing the bending displacement. For the torsional case it induces a larger rotation of the section.

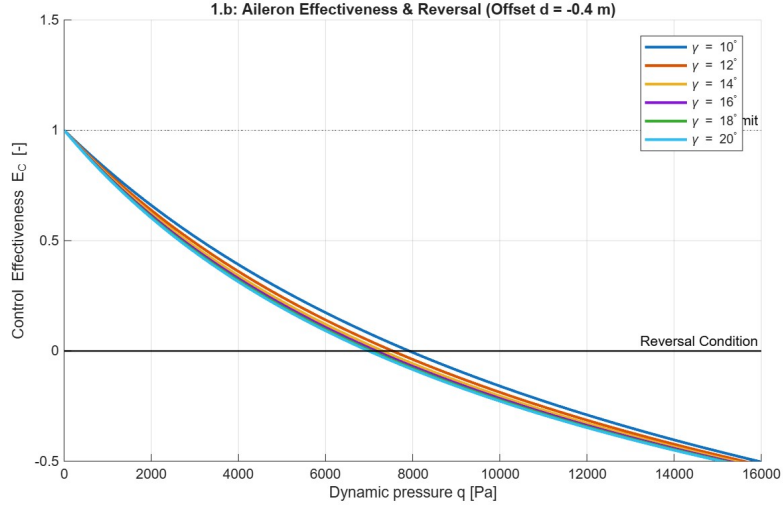


Figure 2: Control effectiveness VS dynamic pressure.

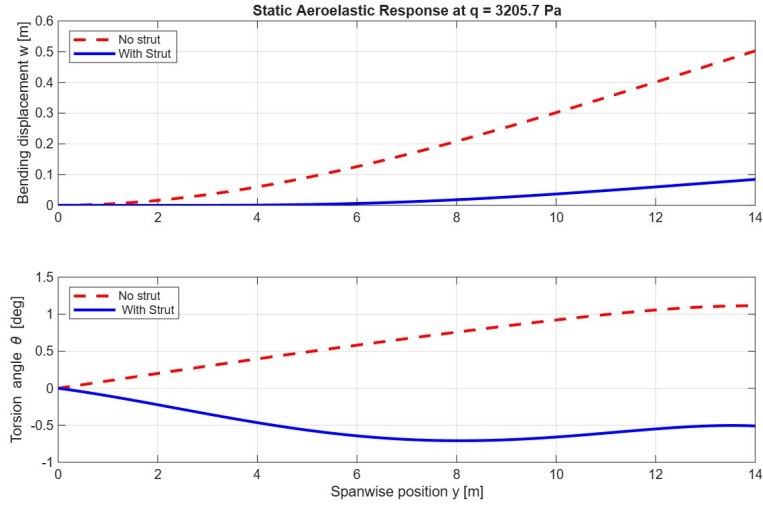


Figure 3: Comparison of spanwise bending (Top) and torsional (Bottom) deformations between the wing with without the strut.

Fourth plot (figure 4) reversal condition is reached only if the strut is positioned between the leading edge and $x \approx -0.18$ m.

The last plot (Figure 5) shows that increasing the number of shape function the solution is converging.

The last request is about modifying the structural design of the wing to achieve an higher value of the divergence dynamic pressure while minimizing the mass increment.

Fixing the chordwise position of the strut, the research of the optimal mass/angle (γ) configuration was performed.

The mass of the wing was considered linearly dependant on the relative stiffness. Instead, for the strut it the mass was considered dependant on the stiffness but also on the length (modifying γ the mass changes).

Therefore, varying one parameter at a time has led the research to an optimal configuration of: $\gamma = 25$ deg, increasing of the wing stiffness by the 40%, halving of the strut stiffness.

After the optimization the mass increased by $\approx 38\%$.

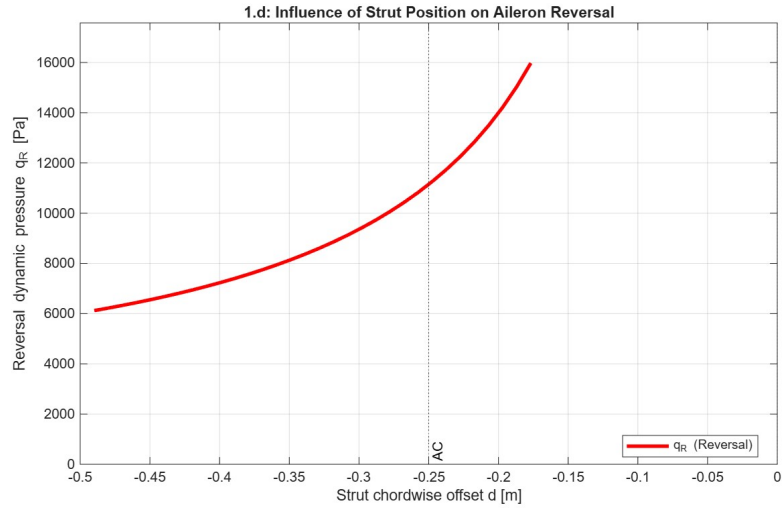


Figure 4: Aileron reversal dynamic pressure as a function of the strut chordwise position.

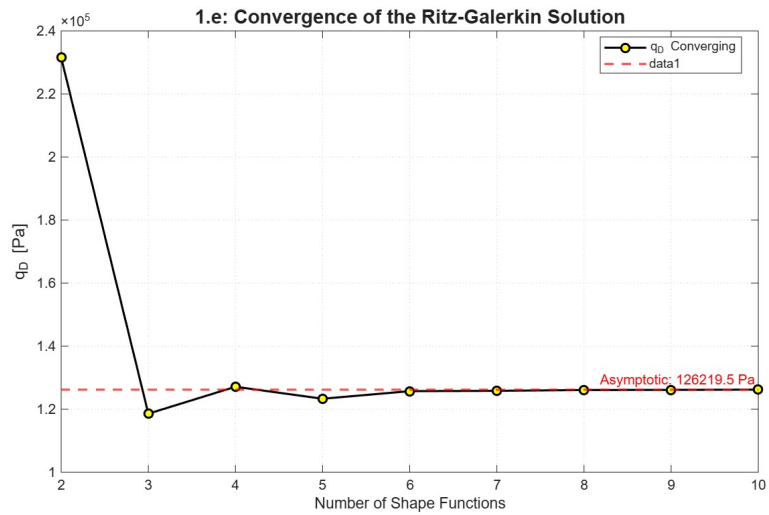


Figure 5: Convergence of results with order of shape function increasing